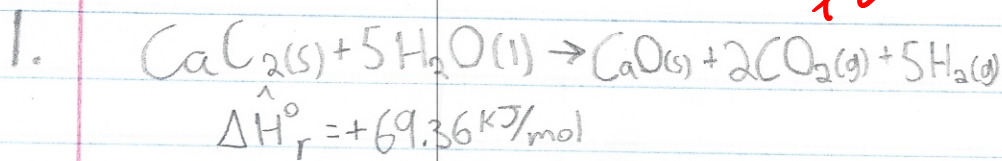


# CHE 210 HW10

Joseph Bildorback

68/72



- a) This reaction is Endothermic at  $25^\circ\text{C}$ , therefore the reactor must be heated to maintain a constant temperature.

If the reactor ran adiabatically, the temperature would lower.

It takes more 

b)  $\Delta \hat{H}_r = \hat{U}_r + P\hat{V} = +69.36 \frac{\text{kJ}}{\text{mol}}$

$PV = nRT$

$\hat{U}_r = 69.36 \frac{\text{kJ}}{\text{mol}} - nRT$

$n = \# \text{ mols in the gas phase} = 7 \text{ mols}$

$\hat{U}_r = 69.36 - 7(8.314 \times 10^{-3})(298.15)$

$T = 25^\circ\text{C} = 298.15 \text{ K}$

$\boxed{\hat{U}_r = 52.01 \text{ kJ}}$

$R = 8.314 \frac{\text{J}}{\text{mol}\cdot\text{K}} \times \frac{1 \text{ kJ}}{1000 \text{ J}} = 8.314 \times 10^{-3} \frac{\text{kJ}}{\text{mol}\cdot\text{K}}$

This is if we assume 1 mol of  $\text{CaC}_2(\text{s})$

$\Delta U =$  Products have higher internal energy than the reactants

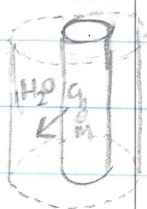
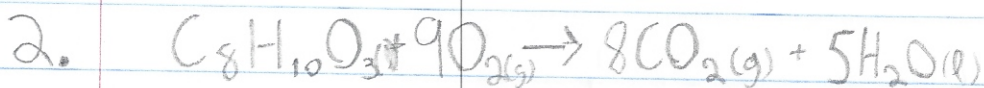
c)

$$\boxed{\begin{array}{l} 150.9 \text{ g} \\ \text{CaC}_2 \end{array}}$$

$$\text{mols CaC}_2 = \frac{150.9}{64.10} = 2.34 \text{ mol}$$

$$\Delta U = q_v = 52 \text{ kJ for 1 mol CaC}_2$$

$$q_v = 52 \times 2.34 = \boxed{121.68 \text{ kJ}}$$



$$\Delta U = Q = m C_v (T_f - T_o)$$

for  $H_2O$ :  $C_v = C_p = 4.18 \text{ J/g}^\circ\text{C}$

$m = 650 \text{ g}$   $T_f = 36.2^\circ\text{C}$   $T_o = 23.6^\circ\text{C}$

$$\Delta U = 650 \text{ g} (4.18 \text{ J/g}^\circ\text{C}) (36.2 - 23.6^\circ\text{C}) = 34234.2 \text{ J}$$

with the inner chamber being the system

$$\Delta U = -34234.2 \text{ J}$$

Since we assume that all of the energy comes from the rxn we can say that

$$\Delta U = n_r \hat{U}_r = -34234 \text{ J} \quad n_r = \frac{2.7 \text{ g}}{154 \text{ g/mol}} = .0175 \text{ g/mol}$$

$$\hat{U}_r = \frac{-34234 \text{ J}}{.0175 \text{ g/mol}} = -1953 \frac{\text{kJ}}{\text{g/mol}}$$

$$\Delta \hat{H}_r = \Delta \hat{U}_r + R T \Delta n_r \quad \Delta n = \text{change of mols in gas phase}$$

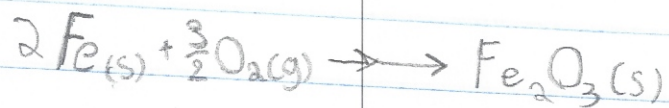
$$-1953 + (8.314 \times 10^{-3}) (298) (-1) = -1955 \frac{\text{kJ}}{\text{g/mol}}$$

$$\Delta \hat{H}_r = \sum \nu_i \Delta \hat{H}_{f,i} = (-1) \Delta \hat{H}_{f,C_8H_{10}O_3} + 8(-393.5) + 5(-285.84) = -1955 \frac{\text{kJ}}{\text{g/mol}}$$

$$\Delta \hat{H}_{f,C_8H_{10}O_3} = -2622 \frac{\text{kJ}}{\text{g/mol}}$$



3. Total kJ:  $100 \frac{\text{kJ}}{\text{h}} \times 6 \text{ h} = 600 \text{ kJ}$



$$\Delta \hat{H}_f \text{Fe}_2\text{O}_3 = \Delta \hat{H}_r = -830.5 \text{ kJ/gmol}$$

$$-830.5 \frac{\text{kJ}}{\text{gmol}} \times n = -600 \text{ kJ} \quad n = .722 \text{ gmols Fe}_2\text{O}_3$$

$$.722 \text{ gmols Fe}_2\text{O}_3 \times \frac{2 \text{ gmol Fe}}{1 \text{ gmol Fe}_2\text{O}_3} = 1.45 \text{ gmol Fe}$$

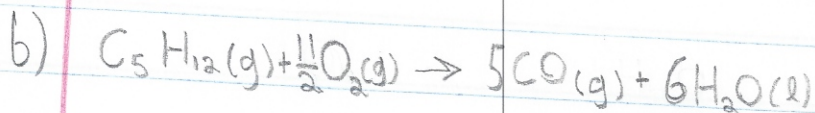
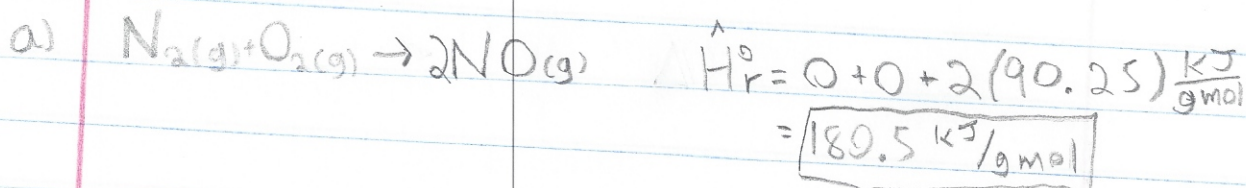
$$M = 1.45 \text{ gmol Fe} \times \frac{56 \text{ g}}{\text{gmol}} = \boxed{81 \text{ g Fe}}$$

$$\Delta M = .722 \text{ gmols Fe}_2\text{O}_3 \times \frac{160 \text{ g}}{\text{gmol}} - 81 \text{ g} = \boxed{35 \text{ g}}$$

$$\% = \frac{81}{100} \times P = 35$$

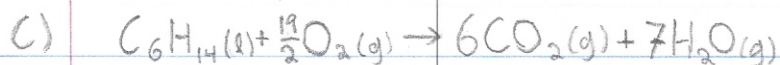
$$\boxed{P = 43\% \text{ increase}}$$

4.



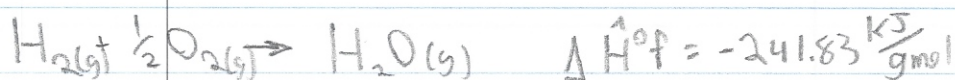
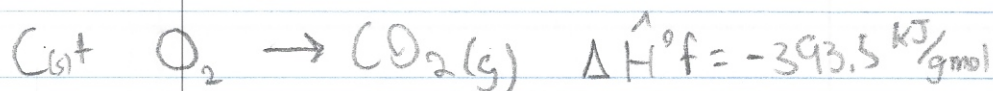
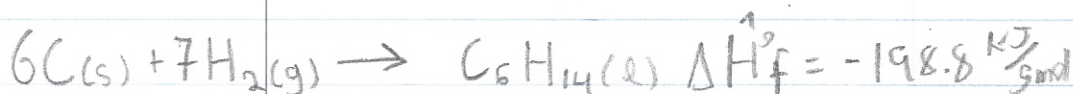
$$\hat{H}_r^\circ = -(-146.76) + 0 + 5(-110.53) + 6(-285.84) \frac{\text{kJ}}{\text{gmol}}$$

$$= \boxed{-2120.93 \text{ kJ/gmol}}$$

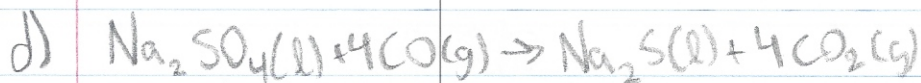


$$\hat{H}_r^\circ = -(-198.8) + 0 + 6(-393.5) + 7(-241.83) \frac{kJ}{gmol}$$

$$= \boxed{-3855 \frac{kJ}{gmol}}$$



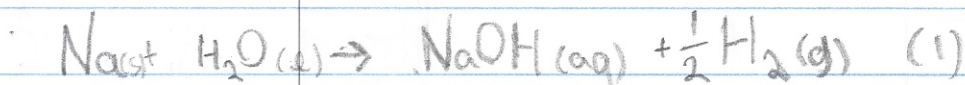
$$-(-198.8) + 6(-393.5) + 7(-241.83) = \boxed{-3855 \frac{kJ}{gmol}}$$



$$\Delta \hat{H}_r^\circ = -373.2 + 6.7 + 4(-393.5) - (-1384.5 + 24.3) - 4(-110.52)$$

$$= \boxed{-138.2 \frac{kJ}{mol}}$$

5.)  $100g Na(s) = 4.35g/mol$   $1mL H_2O = 1g H_2O = 0.056g/mol$



$NaOH$  is also formed, which is a strong base & is a hazardous material.

$$\Delta H_{rxn}^\circ = (-469.15) - (-245.84) = -183.31 \frac{kJ}{g/mol}$$

With this heat of rxn & the  $C_p$  of sodium being  $C_p = 28 \frac{J}{g \cdot mol \cdot ^\circ C}$

$$\Delta H_{rxn} = -183.31 \frac{kJ}{g/mol} \times 0.056g/mol = (4.35g/mol)(28 \frac{J}{g \cdot mol \cdot ^\circ C})(\Delta T)$$

$$\Delta T = 84^\circ C$$

With such a high  $\Delta T$ , it is possible the  $H_2$  gas will ignite

# Comment Summary

Page 1

1. Finish the sentence, though no penalty applied